Review of Simpsons Paradox

Example 1: Data Detective for UC Berkeley

You have just been hired by the UC Berkeley Admissions Committee to investigate a pressing and sensitive issue: gender discrimination in graduate admissions.
Recent statistics suggest that, overall, women appear to have a lower chance of admission compared to men. This finding has raised alarms about potential bias in the admissions process. However, the committee is cautious: they know that simple averages can sometimes be misleading.

They provide you with a dataset that records:

- Applicant gender (Male/Female),
- Department applied to (A–C),
- Admission outcome (1 = admitted, 0 = rejected).

Your task is to carefully analyze the data to discover what is really happening.

1. Are women truly being discriminated against?
2. Or is there a more subtle explanation behind the numbers?

As Berkeley's newly appointed data detective, you must use statistical tools and visualization to uncover the truth.

Example 2: Baseball Batting Averages

Imagine two baseball players, Player A and Player B, competing for the title of "Best Hitter." Looking at the overall batting averages throughout the season, Player A appears to have a higher average than Player B. But when you break it down by half of the season (first half vs. second half), Player B actually has a higher batting average in both halves! How could that be?

Answer: Here is a concrete numerical illustration.

Half	Player A: Hits / At-bats (avg)	Player B: Hits / At-bats (avg)
First half	45/100 = 0.45 (45%)	19/40 = 0.475 (47.5%)
Second half	5/20 = 0.25 (25%)	80/200 = 0.40 (40%)

If we look at the data in each half, we find that B's average is 0.475 vs. A's 0.45. In the second half, B's average is 0.40 vs. A's 0.25. So B is better in both halves. Let's look at overal averages:

Player	Total Hits	Total At-bats	Overall Average
A	50	120	50/120 = 0.417
B	99	240	99/240 = 0.413

Although Player B has the higher batting average in each half individually, Player A appears better overall with 0.417 vs. 0.413 for Player B. This is due to the uneven weighting of at-bats across halves. Player A had most of his at-bats in the half where he did well, whereas Player B had most at-bats in the half where he did worse.

This uneven weighting flips the aggregate comparison, illustrating Simpson's paradox.

Example 3: Kidney Stone Treatment

Doctors are comparing two treatments for kidney stones: Treatment X and Treatment Y. If you look at the overall success rates, Treatment X seems better than Treatment Y. But when doctors separate cases by stone size (small vs. large stones), Treatment Y is better in both categories!

Your task: Please prepare a numerical example similar to excercise 2 to show that most of the patients with large, harder-to-treat stones were given Treatment Y, while Treatment X was mostly used on smaller, easier cases.

This paradox reminds us: the mix of subgroups (like stone size) can completely change the overall picture.